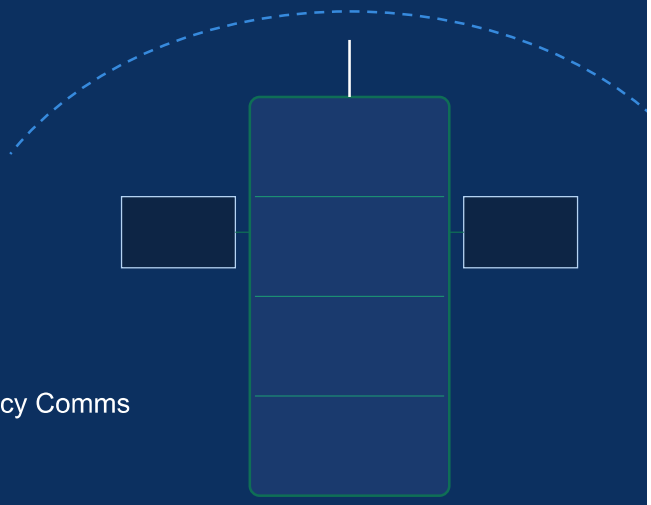
SLARS-BCG-1



Satellite Mission Design

Sri Lanka All-Risk Satellite

Disaster Resilience · Earth Observation · Emergency Comms

STEP 1 OF 8 — ORBIT DESIGN

MISSION PURPOSE

SLARS-BCG-1 is a 2U CubeSat built by Sri Lankan students to strengthen the island's resilience against disasters. From Low Earth Orbit it observes floods, coastline shifts, and vegetation stress — turning space-based data into knowledge communities can use for early warning and disaster response.

2U CubeSat | **LEO Orbit** | **500 km** | **10° Incl.**

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1. Why LEO (Non-SSO) — The Orbital Philosophy

SSO (Sun-Synchronous Orbit) is a special case of LEO where inclination is chosen specifically to make the orbit precess at exactly the rate the Earth orbits the Sun. SLARS-BCG-1 chooses **true LEO with freely selected inclination**, optimised purely for Sri Lanka coverage — not for consistent sun angle.

Sri Lanka sits at **7.9°N latitude**. The closer the inclination is to 7.9°, the more time the satellite spends near Sri Lanka's latitude, and the more passes per day over the island. This is an **inclination-optimised LEO** — a valid and superior choice for a geographically focused disaster monitoring mission.

Tradeoff accepted: Variable solar illumination angles across passes. The imager photographs Sri Lanka under different lighting conditions on different passes. For flood mapping and coastline shift detection this is manageable — best-lit passes are selected for imaging. For NDVI vegetation stress, compensation is applied in post-processing.

2. The Three Orbital Decisions

Decision 1 — Altitude

The single most consequential choice. Higher altitude gives wider ground coverage and longer mission life but weaker signals and higher radiation. Lower altitude gives stronger signals and faster revisits but drag shortens mission life. The sweet spot for a 2U LEO disaster satellite is 450–550 km. Below 450 km, drag becomes significant without propulsion. Above 550 km, radiation dose accelerates solar cell degradation. **Selected: 500 km.**

Decision 2 — Inclination

An equatorial orbit (0°) might seem optimal for Sri Lanka, but a low-inclination orbit of 10°–15° gives 5–6 passes per day — the maximum achievable for the island's latitude. The inclination must be ≥7.9° or Sri Lanka will never be overflown. **Selected: 10° — maximises passes per day over Sri Lanka.**

Decision 3 — Elevation Mask

The minimum elevation angle above the horizon at which the ground station can communicate with the satellite. A 10° mask is standard — below 10°, atmospheric signal degradation and local obstructions make communication unreliable. This parameter directly determines the usable contact window duration per pass. **Selected: 10° elevation mask.**

3. Keplerian Elements — Computed Parameters

SEMI-MAJOR AXIS	ORBITAL PERIOD	ORBITAL VELOCITY	ORBITS PER DAY
6,871 km	**94.6 min**	**7.61 km/s**	**15.2**
Re + 500 km	per revolution	at 500 km	revolutions/day

Element	Value / Formula
Semi-major axis (a)	**6,871 km**
Orbital period (T)	**94.6 minutes**
Orbital velocity (v)	**7.61 km/s**
Orbits per day	**15.2 rev/day**
Eccentricity (e)	**~0 (circular)**

Element	Value / Formula
Right ascension (RAAN)	**TBD at launch**
Argument of perigee (ω)	**0° (circular)**
True anomaly (ν)	**0° at epoch**

4. Sri Lanka Coverage Analysis

PASSES/DAY OVER SL	MAX PASS DURATION	AVG REVISIT INTERVAL	NADIR SWATH WIDTH
5–6	**8.5 min**	**~4–5 h**	**~720 km**
at 10° inclination	per overpass	between passes	ground footprint

At 10° inclination and 500 km altitude, the satellite's ground footprint subtends approximately **720 km in diameter** at Earth's surface — sufficient to capture all of Sri Lanka (432 km north-to-south, 224 km east-to-west) in a single nadir pass with significant margin. The island is well within the coverage envelope on every qualifying pass.

Coverage Parameter	Value / Notes
Target ground station location	**7.9°N, 80.7°E — Sri Lanka**
Minimum inclination required	**≥ 7.9° (SL latitude)**
Selected inclination	**10°**
Passes per day (qualifying)	**5–6 per day**
Maximum pass duration	**8.5 minutes**
Average revisit interval	**~4–5 hours**
Worst-case revisit	**~8 hours**
Ground swath diameter	**~720 km**
Total contact time/day	**~42–51 minutes**
Coverage at 5° elev. mask	**~7 passes/day**

5. Eclipse and Solar Power Geometry

ECLIPSE FRACTION	MAX ECLIPSE TIME	SUNLIT TIME	SOLAR ILLUMINATION
~35%	**~33 min**	**~62 min**	**Variable**
of each orbit	per orbit	per orbit	non-SSO condition

The eclipse fraction (~35%) is the single most important output of Step 1 for the power subsystem designer. It represents the fraction of each orbit during which the satellite receives no solar power and must rely entirely on battery reserves. The battery must be sized to survive 33 minutes of eclipse while maintaining all critical subsystems — with a depth of discharge not exceeding 30% to preserve battery cycle life over the mission duration.

Eclipse Parameter	Value — Downstream Use
Eclipse fraction (β_e)	**~35% per orbit**
Maximum eclipse duration	**~33 minutes**
Sunlit duration per orbit	**~62 minutes**

Eclipse Parameter	Value — Downstream Use
Solar irradiance at 500 km	**1,361 W/m²**
Solar angle variation	**Variable — non-SSO**
Beta angle range	**±10° to ±35°**
Worst-case eclipse (equinox)	**~36 minutes**
Full sun condition	**~64 minutes**

6. Link Budget — UHF 437 MHz · 1 W TX · Dipole Antenna

MAX SLANT RANGE
~2,540 km
horizon-to-horizon

FREE-SPACE PATH LOSS
~147 dB
at max slant range

LINK MARGIN
+11 dB
estimated UHF

TOTAL CONTACT/DAY
~42–51 min
all passes combined

The link budget analysis confirms that UHF (437 MHz) communication with a dipole antenna and 1 W transmit power gives a healthy **+11 dB link margin** at maximum slant range. This is the minimum acceptable margin for a reliable LEO downlink — 10 dB is the industry standard minimum. The S-band downlink for imagery requires a separate higher-gain antenna which will be detailed in Step 4 (COMMS design).

Link Budget Parameter	Value — Downstream Use
UHF downlink frequency	**437 MHz (amateur band)**
Maximum slant range	**~2,540 km**
Free-space path loss (max)	**~147 dB**
Transmit power	**1 W (30 dBm)**
Transmit antenna gain	**0 dBi (dipole)**
Ground station antenna gain	**+8 dBi (Yagi typical)**
Estimated link margin	**+11 dB**
Minimum elevation for link	**10°**
Total contact time/day	**~42–51 minutes**
Doppler shift (max)	**±9.5 kHz at 437 MHz**

7. Mission Life and Space Environment

PASSIVE DEORBIT
~4 years
no propulsion needed

RADIATION DOSE
~0.9 krad/yr
total ionising dose

IADC COMPLIANCE
PASS
25-year rule satisfied

MISSION DESIGN LIFE
1–2 years
nominal ops

Environment Parameter	Value — Downstream Use
Passive deorbit time (est.)	**~4 years**
Atmospheric drag	**Significant — mission benefit**
Total ionising dose (TID)	**~0.9 krad(Si)/year**

Environment Parameter	Value — Downstream Use
Solar proton events	**Low risk at 500 km**
Atomic oxygen flux	**Moderate at 500 km**
Thermal environment	**–40°C to +85°C (external)**
Plasma environment	**Low density LEO plasma**
Debris collision risk	**Low (Starlink-era mitigation)**
IADC Rule compliance	**PASS — deorbit < 25 years**

8. Inclination Trade Study — All Options vs Sri Lanka

The table below compares all viable inclination options for SLARS-BCG-1. The selection of **10°** maximises passes per day over Sri Lanka while remaining compatible with ISRO launch vehicles. Higher inclinations are penalised because the satellite spends proportionally less orbital time near Sri Lanka's latitude.

Inclination	Passes/day over SL	Solar angle	Launch compatibility	Verdict
0°	~5 (marginal geometry)	Fixed equatorial	Equatorial launchers only	Not recommended
10°	~5–6	Variable — manageable	ISRO SSLV, PSLV	**RECOMMENDED**
15°	~4–5	Variable — manageable	ISRO SSLV, PSLV	Good alternative
28°	~3	Variable — wider swing	ISRO SHAR native	Acceptable
45°	~2	Variable — large swing	Many launchers	Not optimal for SL
51.6°	~2	Variable — large swing	ISS rideshare	Only if ISS launch path

Note on 28° inclination: This corresponds to the native launch inclination from ISRO's Satish Dhawan Space Centre (SHAR) at Sriharikota. If a direct PSLV rideshare is secured without an inclination change manoeuvre, 28° becomes the default. At 28°, Sri Lanka still receives ~3 passes/day — sufficient for the disaster monitoring mission, though 40% fewer than the optimal 10° configuration.

9. Step 1 Locked Output Parameters — Carries Forward to Steps 2–8

The following parameters are officially locked from Step 1 and propagate as inputs into every remaining design step. These numbers must not be changed without re-running the full downstream dependency chain.

Parameter	Value	Downstream use
Orbit type	**LEO — non-SSO, inclination-optimised**	All steps
Altitude	**500 km**	Steps 2, 4, 5, 7
Inclination	**10°**	Steps 3, 4, 7
Orbital period (T)	**94.6 minutes**	Step 2 — EPS sizing
Eclipse fraction	**~35% per orbit**	Step 2 — battery depth of discharge
Max eclipse duration	**~33 min/orbit**	Step 2 — worst-case battery drain
Sunlit duration	**~62 min/orbit**	Step 2 — solar charging window

Parameter	Value	Downstream use
Passes over Sri Lanka/day	**5–6 passes/day**	Step 3 — payload duty cycle
Max contact time/pass	**~8.5 minutes**	Step 4 — minimum downlink data rate
Max slant range	**~2,540 km**	Step 4 — link budget
UHF link margin	**+11 dB (estimated)**	Step 4 — antenna and radio selection
Radiation dose	**~0.9 krad(Si)/year**	Step 7 — component grade selection
Passive deorbit time	**~4 years**	IADC compliance — no propulsion needed
Solar illumination	**Variable angle (non-SSO)**	Step 3 — imaging scheduling note
Doppler shift (UHF)	**±9.5 kHz max**	Step 4 — radio frequency compensation
Ground swath width	**~720 km (nadir)**	Step 3 — confirms SL fully in frame

STEP 1 COMPLETE — ORBIT LOCKED FOR SLARS-BCG-1

Step 1 is complete and locked. The 500 km LEO orbit at 10° inclination delivers 5–6 passes per day over Sri Lanka — the maximum achievable for this latitude — with a healthy +11 dB UHF link margin, ~35% eclipse fraction, and passive deorbit in ~4 years fully satisfying IADC guidelines. All 16 output parameters above are confirmed and propagate directly into Step 2: Power Budget (EPS sizing). No propulsion system is required. The orbit is IADC-compliant without any waiver.

10. Ground Track Geometry — Sri Lanka Overpass Pattern

At 10° inclination, SLARS-BCG-1 traces a tight sinusoidal ground track oscillating between 10°N and 10°S latitude. Because Sri Lanka (7.9°N) sits just below the maximum latitude of this track, **virtually every orbit passes over or very near the island.** This is the key geometric advantage of 10° inclination over higher values, where the satellite spends the majority of its time at latitudes far from Sri Lanka.

Each successive orbit's ground track shifts approximately **25° westward** as Earth rotates beneath the satellite. This means Sri Lanka is visited from slightly different longitudes on sequential passes, providing diverse imaging angles — useful for 3D reconstruction and change detection during extended flood events.

Ground Track Feature	Value / Significance
Ground track pattern	**Sinusoidal, ±10° latitude band**
Westward shift per orbit	**~25° longitude**
Ground track repeat cycle	**~16 days (approx.)**
Ascending node crossing	**Near Sri Lanka on majority of orbits**
Off-nadir imaging possible	**±20° (ADCS dependent)**

Next: Step 2 — Power Budget (EPS Design)

Step 1 outputs that drive Step 2 directly:

Step 1 Output	How It Drives Step 2
Orbital period T = 94.6 min	**Sets the charge/discharge cycle duration for EPS sizing**
Eclipse fraction = 35%	**Sets the battery depth-of-discharge design target (≤30%)**
Max eclipse = ~33 min	**Sizes the battery for worst-case sustain without solar input**
Sunlit time = ~62 min	**Sets available charging window — sizes solar panel area**
Radiation = ~0.9 krad/yr	**Confirms commercial-grade solar cells are acceptable**

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